Fabrication Of Gas Sensor

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Introduction

Gas sensors are vital technologies designed to detect and measure the concentration of different gases in the environment. They are used in diverse applications, including air quality monitoring, leak detection, industrial process control, and ensuring personal safety. With ongoing technological advancements, gas sensors are evolving to become more precise, reliable, and cost-effective, paving the way for innovative solutions that contribute to a safer, healthier, and more sustainable world.

Steps in Optical Lithography

1. Substrate Preparation
   Start with a wafer (Si with oxide).
   Dehydration bake at around 110 °C to remove moisture.
   Apply adhesion promoter (HMDS) for better resist adhesion.

2. Photoresist Coating
   Spin coating is used:
      A drop of liquid resist is dispensed.
      Wafer spins at high speed (3000–5000 rpm).
      Uniform thin resist layer forms (thickness depends on spin speed & viscosity).

3. Soft Bake (Pre-bake)
   Bake at 110 °C.
   Removes solvents, hardens resist slightly, reduces flow during exposure.

4. Mask Alignment & Exposure
   Wafer is aligned with a photomask.
   Exposed to UV light (365 nm).

5. Development
   The wafer is dipped in developer solution (e.g., TMAH, NaOH-based).
   In positive resist: exposed areas → dissolve.
   In negative resist: exposed areas → crosslink → stay, unexposed dissolve.

6. Hard Bake (Post-bake)
   Bake at ~120°C.
   Improves adhesion, mechanical strength, and chemical resistance.
7. Inspection
   Optical microscope used to check linewidth, defects, or resist residue.

Application

- Environmental monitoring (air quality, pollution control).
- Industrial safety (leak detection, hazardous gas monitoring).
- Healthcare (breath analysis, medical diagnostics).
- Automotive (emission control, cabin air quality).

Conclusion

Gas sensors are critical technologies that enable air quality monitoring, leak detection, industrial process control, and personal safety. With ongoing advancements, they are becoming more precise, cost-effective, and seamlessly integrated into smart systems and IoT networks. These innovations are driving the development of portable, connected, and intelligent sensing solutions, paving the way for a safer, healthier, and more sustainable future. As their applications expand, gas sensors will continue to play an increasingly vital role in shaping modern life.